investigation," Einstein wrote, "is a clear understanding as to why the 'Schwarzschild singularities' do not exist in physical reality."

As backing for his view, Einstein could also appeal to the internal structure of an idealized star made of matter whose density is constant throughout the stellar interior (right half of Figure 3.5). Such a star was prevented from imploding by the pressure of the gas inside it. Karl Schwarzschild had used general relativity to derive a complete mathematical description of such a star, and his formulas showed that, if one makes the star more and more compact, then in order to counteract the increased strength of its internal gravity, the star's internal pressure must rise higher and higher. As the star's shrinking circumference nears $\frac{9}{8}$ = 1.125 times its critical circumference, Schwarzschild's formulas show the central pressure becoming infinitely large. Since no real gas can ever produce a truly infinite pressure (nor can any other kind of matter), such a star could never get as small as 1.125 times critical, Einstein believed.

Einstein's calculations were correct, but his reading of their message was not. The message he extracted, that no object can ever become as small as the critical circumference, was determined more by Einstein's intuitive opposition to Schwarzschild singularities (black holes) than by the calculations themselves. The correct message, we now know in retrospect, was this:

Einstein's cluster of particles and the constant-density star could never be so compact as to form a black hole because Einstein demanded that some kind of force inside them counterbalance the squeeze of gravity: the force of gas pressure in the case of the star; the centrifugal force due to the particles' motions in the case of the cluster. In fact, it is true that no force whatsoever can resist the squeeze of gravity when an object is very near the critical circumference. But this does not mean the object can never get so small. Rather, it means that, if the object does get that small, then *gravity necessarily overwhelms all other forces inside the object, and squeezes the object into a catastrophic implosion, which forms a black hole.* Since Einstein's calculations did not include the possibility of implosion (he left it out of all his equations), he missed this message.

We are so accustomed to the idea of black holes today that it is hard not to ask, "How could Einstein have been so dumb? How could he leave out the very thing, implosion, that makes black holes?" Such a reaction displays our ignorance of the mindset of nearly *everybody* in the 1920s and 1930s.